

Profile Summary:

Experienced hardware design engineer with good expertise in hardware design, testing, functional safety according to ISO26262 standard to ensure safety and reliability and understanding the V-Cycle. Good hands-on experience with various tools like Medini Analyzer, Mathcad, Mathcad Prime, Simetrix and Matlab . Well experienced in electronic product development, circuit design, verification and validation,etc. Working as a Hardware Design engineer and Functional Safety Engineer (ISO 26262) in Automotive domain.

Overall experience in the Automotive Field: 4 Years 5 months.

Core Competencies

- 4.5 years of experience in the field of Automotive Electronics significantly in Functional safety ISO26262 and Design Analysis.
 - Hands-on experience with simulation and design tools- Mathcad, Mathcad Prime, Simetrix and Matlab.
 - Familiar with automotive communication protocols such as CAN and diagnostic tools (CANoe).
 - Performed worst case analysis of ECU failures during development and validation phases.
 - My experience include majorly working in ISO26262 safety analysis work products **Reliability Analysis, Evaluation of hardware architectural metrics SPFM, LFM, FTA, DFA, DFMEA, Evaluation of safety goal violations due to random hardware failure, Micro pin configuration analysis and Worst Case Analysis(WCA), Sneak circuit Analysis, finding TRD Limits and Risk analysis to propose alternative component.** Electronic
 - Good Knowledge in Electronic Stability Control/Anti-Lock Braking units, Electronic Park Brake system.
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Skills

Experience summary: Product development (industrial, automotive electronics domains), functional safety and design work products, analyzing the faults in Sneak Circuit and Fault injection testing.

Safety: **ISO 26262:** Functional Safety - road vehicles.

ISO 26262: Implementation of ISO 26262 requirements on new / existing products **(including ASIL D safety goals), Safety Engineer** role for **Tier 2**, Worked on ISO 26262 work products **(ISO 26262-5)**, development of **overall safety concept as per target ASIL.**

Work Products: **Reliability Prediction analysis, RHEM** (Random Hardware Fault Metrics), **DFMEA** (Design failure modes and effects analysis), **FTA** (fault tree analysis), **DFA** (dependent failure analysis), **Sneak Circuit Analysis** and **Micro Pin configuration Analysis, Gathering System requirements (SYS_RS) and System architecture design (SYS_AD), Worst case analysis (WCA), Test coverage analysis** (production implementation product) and **CAN Testing.**

Tools: **Ansys Medini, IQ-RM DFMEA, Simetrix, Mathcad, Mathcad Prime, Matlab and IBM**

Doors, knowledge on process related Tool like **PTC Integrity, MS Office.**

Professional Experience

- Working as an Automotive Functional Safety Engineer, Design Analyst since Oct 2021 to present in TCS@Hyderabad.
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Educational Qualifications

- M.Tech in Power Electronics from Gokaraju Rangaraju Institute of Engineering & Technology in 2021 with an CGPA of 8.9.
 - B.Tech in Electrical and Electronics Engineering from Kakatiya University in 2018 with an percentage of 77%.
 - Intermediate in MPC from Prathibha Junior College with an aggregate of 93.4%
 - SSC from ZPHS H with an CGPA of 8.8.
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Work experience / Highlights

1. Project in Detail

Project: Integrated Braking Control (IBC_Gen2)

Tools: ISO 26262 qualified Excel Templates, RHFM and Reliability-IEC62380, RHFM, FTA in Medini , DFA in Excel, DFMEA in IQ-RM, and Configuration Management Tools like PTC Integrity, WCA in Mathcad.

Project Description: Integrated Braking control system is the future braking system (Wet braking) with an advanced feature of ABS, Electronic Park brake, Integrated Trailer Brake Control, 3 Phase BLDC Motor. Implemented functional safety work products as per ISO 26262 part-4 &5.

Role: Played key role in Offshore Team by working on Functional Safety, WCA and analyzing Sneak Circuit Analysis. Received Good feedbacks for both activities

Responsibilities:

- Identify the Safety Relevant and Non-Safety Relevant components.
- Performed **Reliability Analysis** to find out the FIT value for each component
- Perform **RHFM** (Random Hardware Failure metrics) to calculate the SPFM, MPF for each safety goal.
- Perform **Design FMEA** to estimate the Risk Matrix
- Prepare the **Power and Ground Sneak Circuit Test plan** to identify the unexpected path or logic flow within a system which, under certain conditions, can initiate an undesired function or inhibit a desired function.
- Performed **Fault Tree Analysis** for Safety Goals of ASIL levels C and D.
- Functional and durability test case preparations and providing expected range of limits for Voltages, currents and diagnostics.
- Analyzing the circuit and components behavior during **WCA**.
- Performing **Electrical Data DTC Analysis** and identifying the root causes for DTC's that occurred during electrical testing.

2. Projects in Detail

Project: Electro Mechanical Braking

Tools: ISO 26262 qualified Excel Templates, RHFM and Reliability-IEC62380, RHFM, FTA in Medini , DFA in Excel.

Project Description: Electro Mechanical Braking control system is the future under design braking system (Non-Wet braking) with an advanced feature of ABS, 3 Phase BLDC Motor. Implemented functional safety work products as per ISO 26262 part-4 &5.

Role: Played key role in a 3 members Offshore Team by working on Functional Safety like **Random hardware fault metrics, Fault tree analysis**, Design analysis like **WCA, SCA, TRD limits**, component selection.

Personal Profile:

Name : Someshwara Thota
Father Name : Kondaiah Thota
Gender : Male
Marital Status : Married
Languages Known : Telugu, English
Nationality : Indian

Declaration:

I hereby declare that the information furnished above is true to the best of my knowledge.

(Someshwara T)

DATE: 11-Mar-2026

Place: Hyderabad